Detecting intra-categorical structure through contextual bundling on relevant attributes

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Abstract. In large object-attribute matrices, irrelevant attributes distort the derived conceptual orders. Contextual bundling can simplify large binary object-attribute matrices by representing concepts as imperfect representations with a binary model matrix. The degree of correspondence between the model matrix and the original matrix indicates how typical an exemplar is for a concept. It has been shown that contextual bundling can reflect the user's perceived graded category structure. But, this observation is limited to object-attribute matrices that are based on a more abstract set of attributes that relate to the category level. The question arises as to whether contextual bundling reflects the graded category structure even better when the set of attributes becomes more concrete and also more extensive in relation to the exemplar level. In this analysis, we reproduce the existing research results and show that contextual bundling remains robust even without removing outlier exemplars. Since Boolean matrix decomposition, as method of model matrix processing in contextual bundling, is not scalable to larger object-attribute matrices which prevents the compution of more extensive matrizes. We compare Boolean matrix decomposition with more modern approaches that are more efficient. We select the best alternative of Boolean matrix decomposition and analyze whether a graded category structure can be approximated better using concrete object-attribute matrices than on more abstract matrices. Since the more concrete attribute matrices consist of a large number of attributes that can influence each other, we also consider whether selecting only the most important attributes has an impact on the quality of the approximation.

Keywords: category representation · graded category structure · formal context · contextual bundling.

1 Introduction

What is required to describe a concept sufficiently? A fundamental premise of Formal Concept Analysis(FCA) [29] states that the more important an attribute is, the higher it is placed in a hierarchical conceptual order or concept lattice in the terms of FCA. In an earlier study [24], we found that it there is no correspondence between attribute importance and incidence distributions for large binary object attribute matrices, where "large" was specified in terms of the set size of the attribute set. This means by having a large object attribute matrizes as given it forbidden to derive hierarchical concept order. By representing the importance of attributes using user-based attribute weights, we observed that these attribute weights can only be approximated by incidendence distributions if a small number of attributes have been preselected based on well-defined user created attribute weights. The inclusion of irrelevant attributes obviously leads to an additive noise effect [30]. Otherwise, it is not possible to explain why considering only the most important attributes works much better than using all of them, and even a random selection of attributes performs better than using all of them. This phenomenon regarding the ratio between relevant and irrelevant attributes and the separability of data, which we describe for binary object attribute matrices, is already known for continuous data and is described under the term "curse of dimensionality" [21].

In the preceding paragraph, the attribute filtering strategy is used to form sub-contexts from a large binary object-attribute matrix [14]. This strategy is based on the assumption that object-attribute matrices can be described exclusively through attributes. While it is acknowledged that attributes provide a window into the representation of concepts [19], it is no longer assumed that concepts can be described exclusively in terms of attributes. The more common view is that concepts must be described using an intra-categorial structure - the structure within a c ategory - AND an inter-categorial structure - the structure between categories [22, 23]. This alternative strategy involves first grouping incidence data and then deriving a hierarchical conceptual order based on these groupings. Hierarchical Class Modeling (HICLAS) [8, 26] is a method capable of representing both intra- and inter-categorical structures.

The following paper focuses on intra-categorical structuring and examines the extent to which the presence of irrelevant attributes influences their quality. Intracategorical structuring is based on the assumption that concepts are imperfect representations of the world [17]. To account for the imperfect representation in groups of binary incidence data, the idea is to generate a so-called model matrix and compare this model matrix with the original matrix using a certain fit measure. This is the basic idea that we refer to as *contextual bundling*, since we are working with a binary object attribute that is referred to as a *formal context* in the sense of FCA [29, 10], and incorporates simplified groupings that are commonly referred to as *bundles* [26, 6]. Contextual bundling identifies bundles by recognizing rectangular patterns of co-occurring objects and attributes.

The following paper first describes some relevant fundamentals regarding the calculation of similarity in binary object matrices and contextual bundeling. In the subsequent analysis, we first examine whether existing results regarding the quality of intra-categorical structuring on large binary object-attribute matrices by Ceulemans and Storms (2010) [6] can be replicated, and whether it also holds true for more abstract attributes that the presence of irrelevant attributes impairs the quality of contextual bundling. By using the established method of model matrix construction, we can assume a natural bundling and aim to investigate whether more concrete, pre-filtered relevant attributes can outperform contextual bundling for more abstract attributes. We then examine the influence of irrelevant attributes on more concrete attributes in a second analysis. Since such a comparison requires the construction of very large model matrices, we apply a newer image processing method to generate the model matrix.

2 Contextual Bundling

A binary object-attribute matrix or a formal context (in the sense of FCA [10]) $\mathbb{K} := (X, Y, I)$ typically consists of two sets, X (of objects) and Y (of attributes), as well as a binary relation I between them, i.e., $I \subseteq X \times Y$ with $I := \{0, 1\}\dot{I}$f an object x is related to an attribute y via the relation I, this is written as xIy.

2.1 Matrix similarity of formal contexts

Take as given two formal contexts $\mathbb{K}_1 := (X, Y, I)$ and $\mathbb{K}_2(U, V, J)$. For both contexts holds that $X = U$ and $Y = V$ with $I, J := \{0, 1\}$. By associating $I \to J$ or $J \to I$ we got the following subsets [16]:

- $A := I_{\{1\}} \to J_{\{1\}}$ - “equal true”
- $B := I_{\{0\}} \to J_{\{1\}}$ - “false to true”
- $C := I_{\{1\}} \to J_{\{0\}}$ - “true to false”
- $D := I_{\{0\}} \to J_{\{0\}}$ - “equal false”

Same holds true vice versa if $J \to I$.

Properties:

$$|A| + |B| + |C| + |D| = |I| = |J|$$

Jaccard goodness-of-fit index [25] takes Incidences $I = \{0, 1\}$:

$$J = \frac{\mathfrak{n}_{D=1,M=1}}{\mathfrak{n}_{D=1,M=1} + \mathfrak{n}_{D=0,M=1} + \mathfrak{n}_{D=1,M=0}}$$

2.2 Goodness-of-fit and imperfect formal concepts

Jaccard goodness-of-fit index [25] takes Incidences $I = \{0, 1\}$:

$$J = \frac{\mathfrak{n}_{D=1,M=1}}{\mathfrak{n}_{D=1,M=1} + \mathfrak{n}_{D=0,M=1} + \mathfrak{n}_{D=1,M=0}}$$

For computing a model matrix M the following association rule needs to be applied:

$$m_{ij} = \bigoplus_{r}^{R} = e_{ir} f_{ir}$$

Fit measures for model - exemplar matrix comparison

Various fit measures have been proposed to assess the quality of a model matrix relative to an original object-attribute matrix. The Jaccard coefficient J and the S?rensen-Dice coefficient SD are frequently used with binary data to quantify the overlap between observed and modeled occurrences [16]. Let x and y be binary vectors:

$$J(x, y) = \frac{|x \cap y|}{|x \cup y|}, \tag{1}$$

$$SD(x, y) = \frac{2|x \cap y|}{|x| + |y|}. \tag{2}$$

In the following we use Jaccard goodness of fit J as representative fit measure which is usually applied for intra-categorical structuring of concepts [26, 6].

2.3 Model matrix

The model matrix can be described geometrically as follows, in the sense of Belohlavek [4]. An $n \times m$-matrix J is called rectangular if and only if there exist L-sets C in $1, ..., n$:

$$J_{ij} = C(i) \otimes D(j)$$

for $1 \leq n, 1 \geq j \leq m$.

A *rectangular matrix* can be interpreted as a *model matrix* M, since it simplifies the given incidence structure I. For illustration, consider $L = 0, 1$. The fact that M is a model matrix means that entries of M which contains 1s form a rectangular area such as

$$M = \begin{pmatrix} 0\,0\,0\,0\,0\,0\,0 \\ 0\,1\,1\,1\,1\,0\,0 \\ 0\,1\,1\,1\,1\,0\,0 \\ 0\,1\,1\,1\,1\,0\,0 \\ 0\,1\,1\,1\,1\,0\,0 \\ 0\,1\,1\,1\,1\,0\,0 \\ 0\,0\,0\,0\,0\,0\,0 \\ 0\,0\,0\,0\,0\,0\,0 \end{pmatrix}$$

or can be brought into a rectangular area by permutating rows and columns.

2.4 Model matrix construction

The purpose of a model matrix is to capture the structural regularities of a given original matrix using the method for constructing model matrices.

Boolean Matrix Decomposition

Let I be a set of observations in (X, Y, I). Then a model matrix M approximates the data in such a way that the following loss function is minimized [8, 26, 6]:

$$L = \sum_{i=1}^{O} \sum_{j=1}^{A} (d_{ij} - m_{ij})^2$$

Subject to the restriction that, for a permutation of the objects $o \in O$ and the attributes $a \in A$ R, the model matrix M may contain overlapping rectangles of ones. This approximation is achieved by reducing the objects and attributes to overlapping clusters, known as *bundles*, on R using an alternating least squares method or simulated annealing [7]

Boolean Matrix Factorization (BMF) as Image processing

Structural analysis approaches and Boolean matrix decomposition break down an instance-attribute matrix into a set of binary factors that approximate the original data. These methods are closely related to Boolean matrix decomposition, which is used in various fields, including image processing. Trnečka and Trnečková proposed interpreting Boolean matrices as binary images and applying image processing techniques to identify dense rectangular regions corresponding to the factors [27]. When a binary object attribute matrix groups objects based on shared attributes, humans are able to easily identify clusters as compact rectangles, even if they are not "perfect rectangles." Existing BMF algorithms operate on the same principle but are unable to identify natural factors as humans can; for this reason, more flexible image processing methods have been proposed instead.

In image processing, a binary object attribute matrix $\mathbb{K} = (X, Y, I)$ of size $m \times n$ pixels is specified, where $m = |X|$ and $n = |Y|$. To identify natural patterns in the data, the matrix must be permuted to reveal a banded structure (or a staircase pattern) [11].

Trnecka and Trneckova (2025) [27] compared three representative methods for computing a banded structure:

- Spectral ordering [2]
- Barycenter [18]
- the Alternating [12]

After creating a banded structure, rectangular patterns within the re-ordered matrix must be identified using an image processing method. Trnecka and Trneckova (2025) [27] compared the following:

- Connected Components [13]: groups pixels as connected components based on a connectivity criterion (e.g., neighborhood 4-neighborhood or 8-neighborhood). Only incidences $i := 1$ next to each other in the vertical or horizontal direction will be connected.
- Simple Linear Clustering (SLIC) [1]: based on k-means clustering spatial consistent clusters or superpixels are identified.1
- (modified) Quad Tree [13]: the image is recursively divided into quadrants until a stop criterion is reached. Originally Quad Tree assumes a symmetric division on processing rows and columns. In a modified version the splitting point was chosen which creates split parts as different as possible.
- OTSU [20]: bundling based on between-class variance
-
- Spatial Filtering [13]: by applying a filter (or mask1) pixel values are modified based on the spatial relationship with neighborhood pixels. Applied method keeps focus on median filtering which is a linear spatial filter based on weighted sum.1

2.5 Contextual Bundling

Taking a original matrix $\mathbb{K}_O := (X, Y, I)$ and model matrix $\mathbb{K}_M := (X, Y, M)$ consisting of L-sets C. Contextual bundling describes each object element $x \in X$ and each attribute element $y \in Y$ in terms of a fit measure J i.e. $j_x \in J_X$ and $j_y \in J_Y$.

3 Outline

When describing categories in terms of their intra-categorical structure, it is assumed that a graded category structure exists. The idea of a graded category structure was first introduced by Rosch and Mervis (1975) [22], who state: "We know that a chair is a more appropriate example of the category *furniture* than a radio, and that some chairs correspond better to our conception or image of a chair than others." [22] The typicality of elements in the category, expressed through typicality ratings, captures the general intuition that some exemplars are more representative than others. A typicality gradient is assumed, according to which some exemplars (exemplars: object) are more typical or representative of the category (category: set of objects) than others (e.g., [28, 9]). According to prototype theory, the typicality of category exemplars is based on the extent to which they possess attributes that are characteristic of the concept (i.e., based on the similarity of the exemplars to the category prototype or to other exemplars). *Family resemblance*, as the fundamental principle of prototype theory, already assumes that typicity refers to a distribution structure of attributes, while typical attributes of a category form data clusters: "The attribute that occurs most frequently among the members of a category and least frequently among the members of contrasting categories is, by definition, the most valid indicator of membership in the category in question." [22] Contextual bundling takes up the idea of family resemblance and describes concepts (or categories in the sense of category representation) as imperfect representations through the creation of a model matrix.

There is already evidence that contextual bundling enables the approximation of a graded category structure [26, 6]. In particular, Ceulemans and Storms (2010) [6] were the first to apply contextual bundling to large binary object-attribute matrices and to demonstrate that the perceived hierarchical category structure, as reflected in typicity ratings, correlates well with the intra-categorical fit of contextual bundles. Although the concept of contextual bundling sounds intuitive, previous research findings are limited because the influence of the relationship between relevant and irrelevant attributes has not yet been investigated. Furthermore, the analysis deals only with incidence data for more abstract attributes and disregards incidence data for more concrete attributes. Boolean matrix decomposition, as an existing method for constructing model matrices, prevents computations on very large binary object-attribute matrices because it is too inefficient. However, newer methods for constructing model matrices based on Boolean matrix factorization and image processing are significantly more efficient and can be applied to binary object-attribute matrices of any size [27]. Taking this more modern method for constructing model matrices into account, we aim to investigate whether the quality of contextual bundling depends on the ratio of relevant to irrelevant attributes, as revealed in the context of more abstract and even more concrete attribute sets.

Based on our previous observation [24], that an inter-categorical structure can be better represented using concrete attributes than abstract ones, provided that certain attributes are filtered out, we assume that this also applies to the representation of graded-category structures.

4 Method

We measure the extent to which the graded category structure is reflected in the incidence structure of a context bundle. Typicality ratings are collected empirically by asking participants to evaluate how well a given example represents a category; averaging these judgments across many raters yields a continuous typicality score, usually ranging from 1 to 20 [9], which reflects the perceived centrality within the category.

The Leuven Conceptual Data set both extensive, human-created binary object-attribute matrices and human-assigned typicity ratings [9]. The Leuven Conceptual Data consist of 16 categories, with 5 categories (i.e., Birds, Fish, Insects, Mammals, Reptiles) grouped under Animals and 6 categories (i.e., Clothing, Kitchen utensils, Musical instruments, Tools, Vehicles, Weapons) have been grouped under Artifacts. Respondents were asked to list either attributes for a specific category (so-called category attributes or category-level) or for a specific exemplar (exemplar attributes or exemplar level). These two types of attributes differ from one another (for example: category attributes identified for the Fish category overlap with the union of the exemplar attributes identified for the respective exemplar in that category). The result is two versions of the applicability matrices: 'Exemplar by Category Attribute' matrices and 'Exemplar by Category Attribute' matrices. Each applicability matrix was completed separately by four study participants (i.e., the participants indicated whether objects possessed certain attributes). To obtain a single binary matrix from these four matrices, at least two participants had to agree in order to disqualify an item under the majority rule [3]. The typicality ratings range from 1 to 20, as they are calculated as averages of the scores assigned by respondents on a scale of twenty items.

In terms of the set of objects `objects`, the set of attributes `attributes`, and the incidence density `density`, the aggregated categories Animals and Artifacts can be described in the Leuven Concept Data as follows (taken from [3]):

Table 1: Leuven Conceptual Data (overview) of the categories Animals and Artifacts

Category	level	objects	attributes	density
Animals	categories	129	225	.32
Animals	exemplar	129	764	.13
Artifacts	categories	166	301	.23
Artifacts	exemplar	166	1295	.09

We aim to investigate whether the Jaccard goodness of fit coefficient at the object level J^O) is capable of reflecting human-assigned typicality ratings per category, which are based on graded category structures. To this end, the Jaccard goodness of fit J^O) was calculated for the contextual bundles of the aggregated categories Animals and Artifacts. In addition to the matrices at the category and exemplar levels, we created exemplar-filtered matrices that are the same size as the matrices at the category level. As in our previous analysis, we used two user-based measures to weight attributes - namely, the feature generation frequencies and feature importance ratings - as well as random filtering. Further details on these methods can also be found in [24]. Since feature importance ratings are only available for each individual category and is not aggregated for Animals and Artifacts, we filtered the attributes for each category and then selected these filtered attributes for the aggregated categories Animals and Artifacts, respectively.

5 Analysis 1

In this section, we investigate whether the induced context bundles capture the graded intra-categorical structure indicated by the Jaccard goodness of fit at the object level J^O and consistent with human judgments based on Boolean matrix decomposition. First, we reproduce the results reported by Ceulemans and Storms (2010) [6] at the category level regarding the evaluation of context bundles. Next, we consider Animals and Artifacts at the exemplar level with filtered attributes and measure model fit as well as its correlation with typicity ratings.

5.1 Reproduction of results by Ceulemans and Storms (2010)

Ceulemans and Storms (2010) examined only the aggregated domains Animals and Artifacts, which appear at the category level in the Leuven conceptual data. As in their study, the correlation analysis is based on the calculation of the Pearson correlation coefficient, or $\tilde{\rho}$, and the significance values p. The correlation values p are reported using the observed significance levels $* : p < .05, ** : p < .01, *** : p < .001$. For specific reasons, the results already obtained include some omissions in the object set for animals and artifacts. Additionally, the results for certain categories of Artifacts were not reported.

The following table 2 summarizes all correlations between typicality rating and Jaccard goodness of fit at the object level J^O for individual Animal categories (Birds, Fish, Insects, Mammals, and Reptiles). First, we list the results obtained by Ceulemans and Storms in `old (excl)`, taking into account the exclusions made. In comparison, we present our reproduced correlations taking into account the same object exclusions `new (excl)` as well as the correlations with the complete object set without exclusions `new (no-excl)`.

Table 2: Animal-domain category-level correlation results by Ceulemans and Storms (2010) reproduced (boolean matrix decomposition)

Category	old (excl)	new (excl)	new (no-excl)
Birds	.66**	.686**	.686**
Fish	.57**	.605*	.765**
Insects	.43*	.654**	.654**
Mammals	.04	.053	.053
Reptiles	.51*	.452*	.452*

For Animals, the present results exclude only three exemplar among the Fish and show moderate correlation coefficients for Birds (0.66), Fish (0.57), and Reptiles (0.51). The correlations determined for Reptiles (0.51) were weak. The correlations in all these categories were significant to highly significant, with the exception of Mammals (0.04). The reproduced correlation results are generally close to the original ones. We obtain significant results for the same categories (Birds, Fish, Insects, and Reptiles) and no significant results for Mammals. The correlation strength differs only slightly among Birds (0.686), Fish (0.605), Mammals (0.053), and Reptiles (0.452). For Insects (0.654), we observe better results than originally intended. When the correlation analysis is performed on the complete dataset without exclusions, we find that the hierarchical category structure for Fish (0.765) can be better captured.

In addition, Ceulemans and Storms analyzed the correlation for Artifacts (Clothing, Kitchen utensils, Musical instruments, Tools, Vehicles, and Weapons) included in the Leuven conceptual data. The following table 3 shows their results, as well as our reproduction of them in the same manner as we did previously for the Animals.

Table 3: Artifact-domain category-level correlation results by Ceulemans and Storms (2010) reproduced (boolean matrix decomposition)

Category	old (excl)	new (excl)	new (no-excl)
Clothing	.64**	.67***	.67***
Kitchen utensils	–	.17	.16
Musical instruments	–	.22	.23
Tools	.60*	.47*	.33
Vehicles	–	.45*	.60**
Weapons	–	-.40	-.27

Ceulemans and Storms found moderate correlations for Artifacts in the categories of Clothing (0.64) and Tools (0.60) - both of which were significant. Results for other categories (Kitchen utensils, Musical instruments, Vehicles, and Weapons) were not reported. Our replicated results show roughly the same degree of correlation strength for Clothing (0.67, highly significant), but only a weak correlation for Tools (0.47, significant). In this context, we observe only moderate correlations for Clothing, weak correlations for Tools, Vehicles (0.45, significant), and Weapons (0.40, not significant), as well as very weak results for Kitchen utensils (0.17, not significant) and Musical instruments (0.22, not significant). Regarding Artifacts, Ceulemans and Storms excluded a number of objects. Without these exclusions, the correlation for Tools is significantly weaker (.33) and no longer significant, while the correlation for Vehicles becomes moderate (.60).

Overall, we can conclude that we have successfully reproduced Ceulemans and Storms' results at the category level. Our results suggest that the proposed exclusion objects offer no significant advantage, so we conduct the following analyses without any reduction. As originally noted, the graded category structure can be observed in Animals to a certain extent. The overall picture for Artifacts shows that the graded structure at the category level, based on Boolean matrix decomposition, is only inadequately captured.

5.2 Model fit

In the following, we investigate whether the models generated by a Boolean matrix decomposition fit an original binary object-attribute matrix. To do this, we calculate a matrix-wide Jaccard goodness of fit for the model matrices J^Mİn Boolean matrix decomposition, the number of clusters to be generated must be specified. In each case, we set a cluster size of 5, as suggested by Ceulemans and Storms [6]. We perform this analysis at the category level and compare the results with filtered exemplar matrices for Animals and Artifacts that are identical in terms of the size of the attribute set and the size of the object set. The following table 4 summarizes the model fit values for the category level `cat`, the `exe (rnd)` filtered at the exemplar level by random attributes, the `exe (frq)`, and the `exe (imp)` filtered by feature importance ratings. We interpret the model fit values as follows: ≥ 0.9 is very strong, $0.7 \leq 0.9$ is strong, $0.5 \leq 0.7$ is moderate, $0.3 \leq 0.5$ is weak, and < 0.3 is very weak.

Table 4: Model fit correlation analysis based on Boolean Matrix decomposition

Category	cat	exe (rnd)	exe (frq)	exe (imp)
Animals	.757	.156	.146	.214
Artifacts	.704	.134	.137	.174

The results indicate that, at the category level, there is good fit between the model matrix and the original matrix for Animals and Artifacts. Model matrices at the examplar level, generated based on filtered attributes, result in a model that does not match the original matrix. Randomly filtered attributes for Animals (0.156) and Artifacts (0.134), feature generation frequencies for filtered attributes in Animals (0.146) and Artifacts (0.137), as well as feature importance ratings for filtered attributes in Animals (0.214) and Artifacts (0.174) - are very weak in every case.

5.3 Correlation analysis

In the following list, we summarize key correlations that we were able to identify not only at the category level but also for object-attribute matrices filtered by attributes. Negative correlation values indicate that objects in a category interpreted as typical exhibit a high Jaccard goodness of fit at the object level J^O.

– Animals

  - `cat`: .69*** (Birds), .765*** (Fish), .65*** (Insects), .45* (Reptiles)
  - `exe (rnd)`: -.48** (Fish), .53* (Reptiles)
  - `exe (frq)`: .67*** (Insects), -.57** (Birds)
  - `exe (imp)`: .58** (Birds), -.525* (Fish)
- Artifacts
  - `cat`: .598*** (Vehicles), .672*** (Clothing)
  - `exe (rnd)`: no significant results
  - `exe (frq)`: .539* (Weapons)
  - `exe (imp)`: .699*** (Vehicles)

In addition to the correlation values previously observed for categories (see above), the matrices filtered by traits do not reflect the graded category structure. Model matrices yield implausible negative correlation results - particularly for Animals in random attribute filtering (Fish -0.48 and Reptiles 0.53), in attribute filtering based on feature generation frequencies (Insects -0.67 and Birds -0.57) and in attribute filtering based on trait significance ratings (Insects 0.58 and Birds -0.525) - in each case significant or greater. The model matrices for Artifacts, on the other hand, appear to capture the graded category structure only very incompletely. Attribute filtering based on feature generation frequencies shows a significant correlation only for Weapons (0.539), and attribute filtering based on feature importance ratings for Vehicles (0.699).

6 Analysis 2

In Analysis 1, we calculated the model matrix based on a Boolean matrix decomposition that cannot be scaled to larger object attribute matrices, such as the matrices for Animals and Artifacts at the exemplar level as part of the Leuven Conceptual Data. To enable the creation of the model matrix for Animals and Artifacts at the exemplar level as well, we use an image processing pipeline for Boolean matrix factorization. Based on the selected banded structure method Barycenter and the image processing method OTSU, context bundles are projected onto the original incidence structure of the binary object attribute matrix, resulting in the model matrix. As in the analysis, we compare the results with the model construction at the category level and the attribute-filtered matrices.

6.1 Model fit

Within a range of plausible parameters, we select the model with the highest matrix-wide Jaccard similarity index J^M. By selecting the best models for each run, we obtain the following results 5 for model fit, compared at the category level `cat`, at the full exemplar level `exe`, and filtered exemplar levels. As before, we filter attributes randomly `exe (rnd)`, by feature generation frequencies `exe (frq)`, and by feature importance ratings `exe (imp)`.

Table 5: Model fit based on Boolean Matrix factorization (image processing)

Category	cat	exe	exe (rnd)	exe (frq)	exe (imp)
Animals	.450	.122	.359	.143	.313
Artifacts	.262	.096	.228	.265	.227

By incorporating the image processing pipeline into the creation of the model matrix, we generally observe a poorer fit between the model matrix and the original matrix. The category level for Animals fits slightly better in some cases - though only marginally for the category level (.450), the exemplar-level filtered by random attributes (.359), and the feature importance filtering (.313).

6.2 Correlation analysis

As in Analysis 1, we list significant correlations by category at the category level `cat`, at the exemplar level `exe`, filtered by random values `exe (rnd)`, and filtered by the feature generation frequencies `exe (frq)` and filtered by the feature importance ratings `exe (imp)`:

- Animals
  - `cat`: .68*** (Fish), .67** (Reptiles)
  - `exe`: .37* (Birds), .49* (Fish), -.66** (Reptiles)
  - `exe (rnd)`: .39* (Birds)
  - `exe (frq)`: .55** (Birds), .46* (Insects), .5** (Mammals)
  - `exe (imp)`: -.5** (Birds), -.57 (Fish)
- Artifacts
  - `cat`: -.64*** (Tools), .48** (Vehicles), -.55* (Weapons)
  - `exe`: -.73*** (Weapons)
  - `exe (rnd)`: .37* (Vehicles)
  - `exe (frq)`: -.46** (Kitchen utensils), -.48* (Weapons)
  - `exe (imp)`: .45* (Musical instruments)

Consistent with the poorer model fit, we observe several inconsistencies regarding the graded category structure - specifically, at the exemplar level for Animals in the case of Reptiles (-0.66), at the exemplar level filtered by feature importance ratings for Birds (-0.5) and Fish (-0.57); as well as for Artifacts at the category level for Tools (-0.64) and Weapons (-0.55), at the exemplar level for Weapons (-0.73), and at the exemplar level for attributes filtered by feature generation frequencies for Kitchen utensils (-0.46) and Weapons. Only for Animals at the category level can a moderate correlation be observed for Fish (0.68) and Reptiles (0.67). This suggests that a graded category structure applies only partially in specific cases, namely for Animals at the category level. The suitability in this specific case aligns with our observation within the framework of Boolean matrix decomposition.

7 Discussion

In this contribution, we have examined the influence of irrelevant attributes on the quality of intra-category structuring. Following the approach proposed by HICLAS [8], we group incidence data by calculating a model matrix and interpret the intra-category structure as the correspondence between the model matrix and the original matrix on object level basis within a calculated clusters - referred to as a contextual bundle. To ensure comparability with our reference work by Ceulemans and Storms (2010) [6], we selected Leuven Conceptual Data [9]. In addition to the classical method of Boolean matrix decomposition, we were able to generate and analyze model matrices at the exemplar level for the first time using an image processing pipeline for Boolean matrix factorization consisting of Barycenter [18] and OTSU [20]. Consistent with Ceulemans and Storms (2010) [6], we used the Jaccard goodness of fit [15] as a quality measure.

Limitations: The research design described focuses on the effects of irrelevant attributes and therefore does not take into account other existing or even better approaches in the field of contextual bundling. Other common methods for calculating banded structure and for image processing were listed to compute a model matrix [27]. Since the range of existing image processing methods is broad, it can be assumed that additional methods will be considered in the future. A quantitative evaluation of the image processing used to create the model matrix was not included. As an alternative to the Jaccard goodness-of-fit measure, there are many other similarity measures that may outperform it in some cases. In particular, the alternative measure proposed by Rosch and Mervis (1975) [22] appears to be more robust [3]. It has been proposed to identify the intra-categorical structure without requiring context bundles [3]. This alternative strategy determines exclusively a graded category structure based on a given binary object-attribute matrix. Nevertheless, such a strategy depends on incidences of other attributes, which we do not wish to consider, as we assume that most attributes become irrelevant and distort our results.

The question of whether there are intra-categorical structures and according to which principles they function was largely raised by Rosch and his colleagues [22, 23]. Whether these intra-categorical structures can be interpreted more strictly as formal concepts or more loosely through the use of contextual bundling remains an open question. Belohlavek and Mikula (2024) have already pointed this out, explaining: “The absence of a single attribute on a given object may result in excluding the object from the scope of a given formal concept.” [5] Although their work suggests that groups in incidence data can be interpreted as formal concepts, their quantitative results also indicate that formal concepts cannot generally be identified. For this reason, a looser interpretation of such groupings in the sense of contextual bundling rather than formal concepts appears more plausible.

What do we observe?

Analysis 1: We were able to reproduce and expand upon the results reported by Ceulemans and Storms (2010) [6]. It became clear that the model developed at the category level generally fits the original matrix well, but that the graded category structure can only be reflected to a certain extent in the case of Animals. The simultaneous consideration of abstract attributes at the category level alongside concrete attributes at the exemplar level, as well as their filtered variants, made it clear that contextual bundling exclusively at the category level yields useful results for Animals. In contrast to a purely inter-categorical approach, as is customary in FCA [24], no advantage from attribute filtering was observed in the intra-categorical analysis.

Analysis 2: The results suggest that the image-processing approach generally leads to poorer model fit and inconsistent results with regard to the graded category structure. Only at the category level for Animals were we able to achieve a plausible model that fits to a certain extent and partially captures the graded category structure. Even in this case, attribute filtering does not provide any further significant advantage.

8 Conclusion

The results of this analysis are relevant in two respects. First, they show that attribute filtering is not improved by taking into account a hierarchical category structure of concepts. Attribute filtering appears to be helpful only in a simplified scenario in which a specific category is fixed. Second, it becomes clear that contextual bundling depends on a well-fitting model matrix, whereas this is actually only observed at the category level in animals that are evidently represented by a suitable incidence structure.

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